CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Disk

Lecture No.- 1



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Recap of Previous Lecture



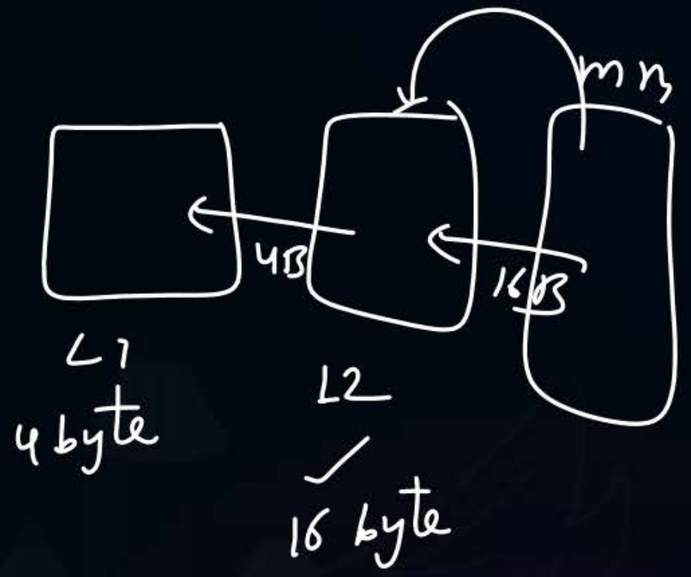
Topic — Multilevel Cache

Topic — Cache Inclusion Policy

Topics to be Covered

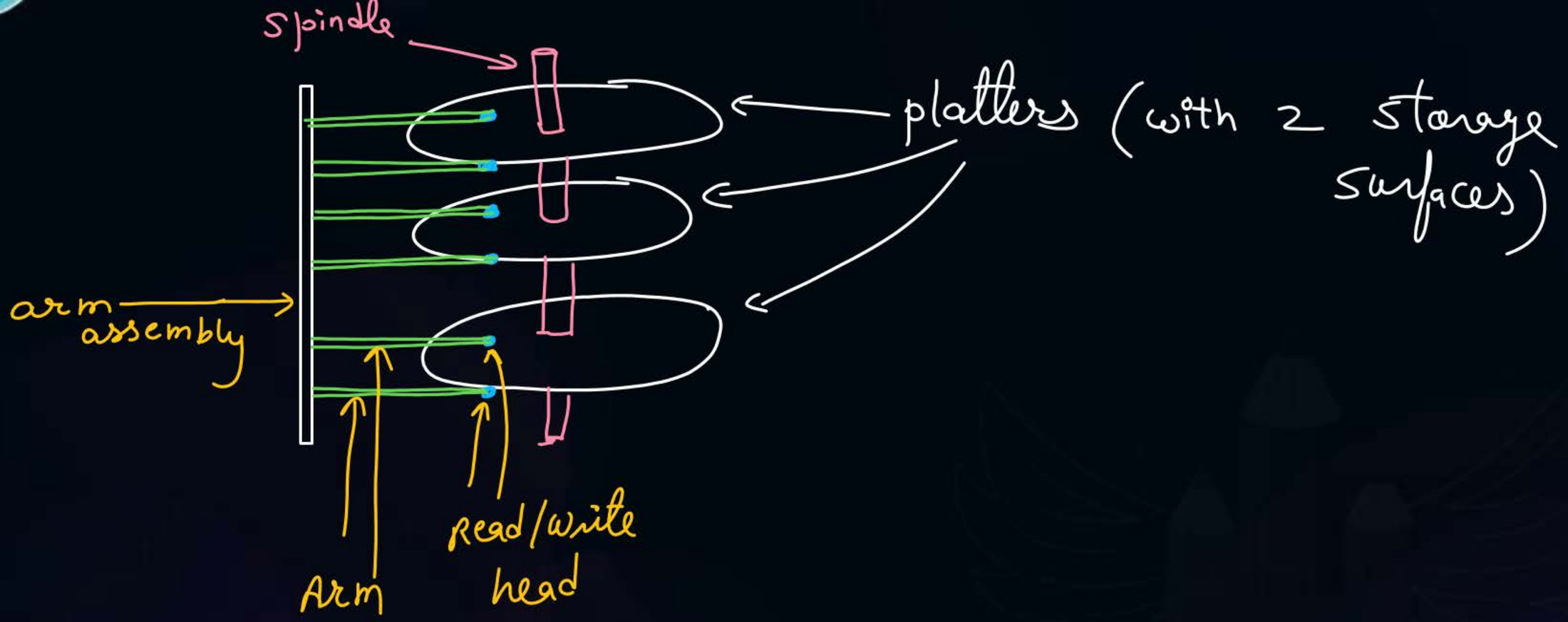


- **Topic** Magnetic Disk
- **Topic** Disk Capacity
- **Topic** Disk Access Time





Topic : Magnetic Disk





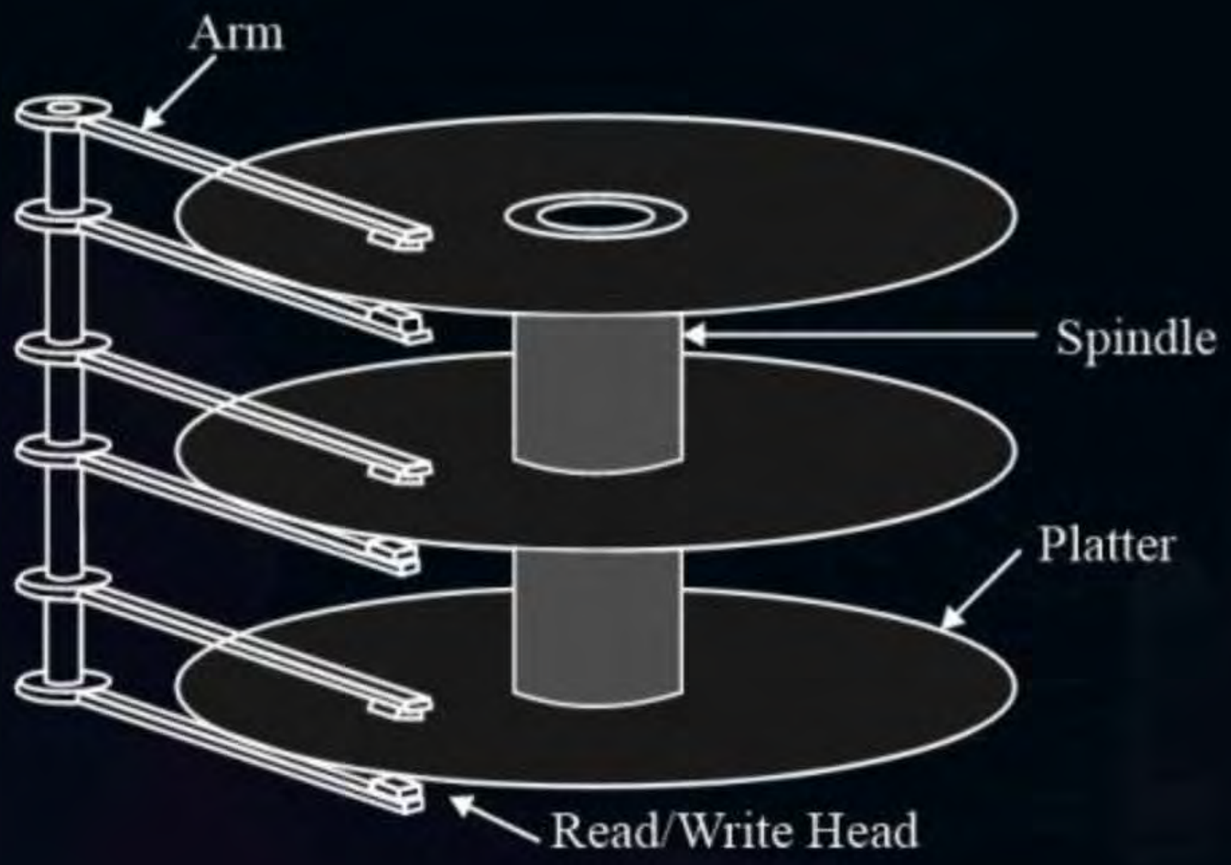
Topic : Magnetic Disk





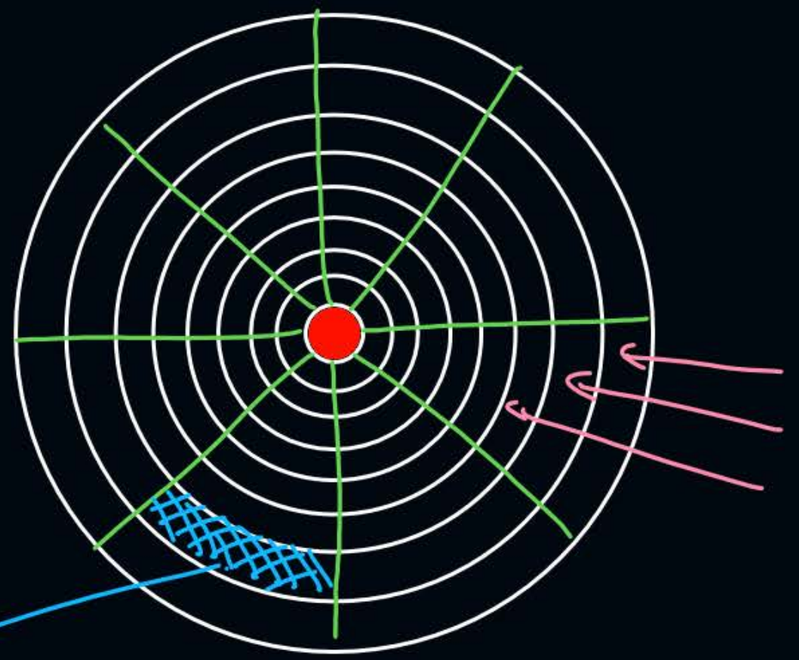


Topic : Magnetic Disk



Top view of disk:-

all the surfaces are divided into equidistance tracks.



sector is the smallest unit of the disk which can be read or written at once.

Each sector gets an address.



Topic : Magnetic Disk

Number of surfaces in disk: 2 * # platters

No. of tracks in disk = no. of surfaces of disk * no. of tracks per surface

no. of sectors on disk = # of tracks in disk * no. of sectors per track



Topic : Sector Capacity

✓

Constant sector capacity

or

Variable sector density

or

Constant angular velocity

×

Variable sector capacity

or

Constant sector density

or

Constant linear velocity



Topic : Magnetic Disk

Number of surfaces in disk:

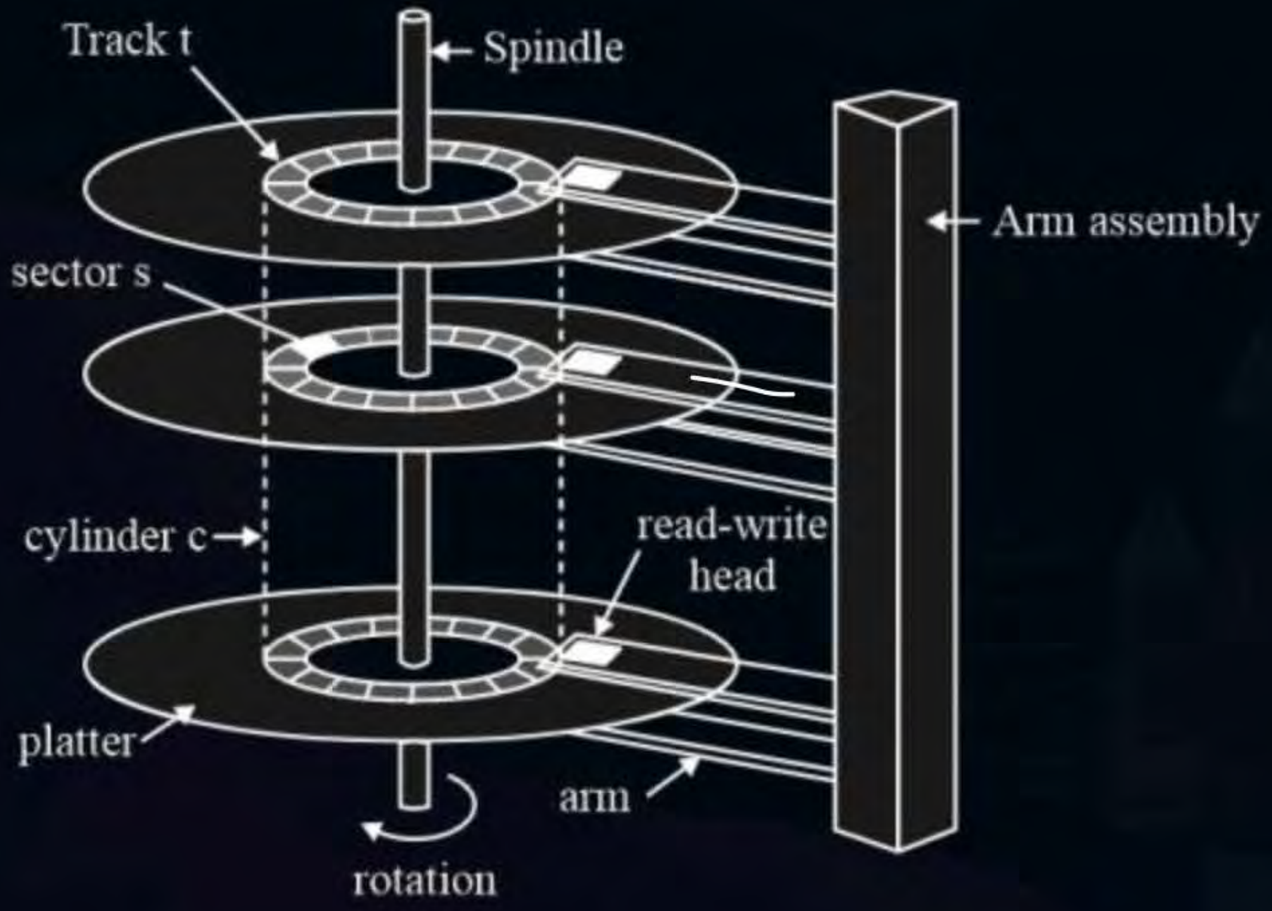
Number of tracks on disk:

Number of sectors in disk:

Number of bytes on disk: no. of sectors on disk * per sector capacity



Topic : Magnetic Disk



[NAT]

#Q. Consider a disk with 32 platters each with 2 recording surfaces. There are 128 tracks per surface and 32 sectors per track. Each sector has equal capacity of 1KBytes.

Calculate:

1. Number of surfaces in disk: $32 * 2 = 64$
2. Number of tracks on disk: $64 * 128 = 2^{13}$
3. Number of sectors in disk: $2^{13} * 32 = 2^{18}$
4. Number of bytes on disk: $2^{18} * 1KB = 2^{28} B = 256MB$
5. Number of bits for disk addressing: 18 bits

Disk access time = seek time + rotational latency + 1 sector transfer time + additional delay

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Note :- In one rotation, entire track can be transferred.



Topic : Disk Access Time

Seek Time: Time required to position the arm over the desired track

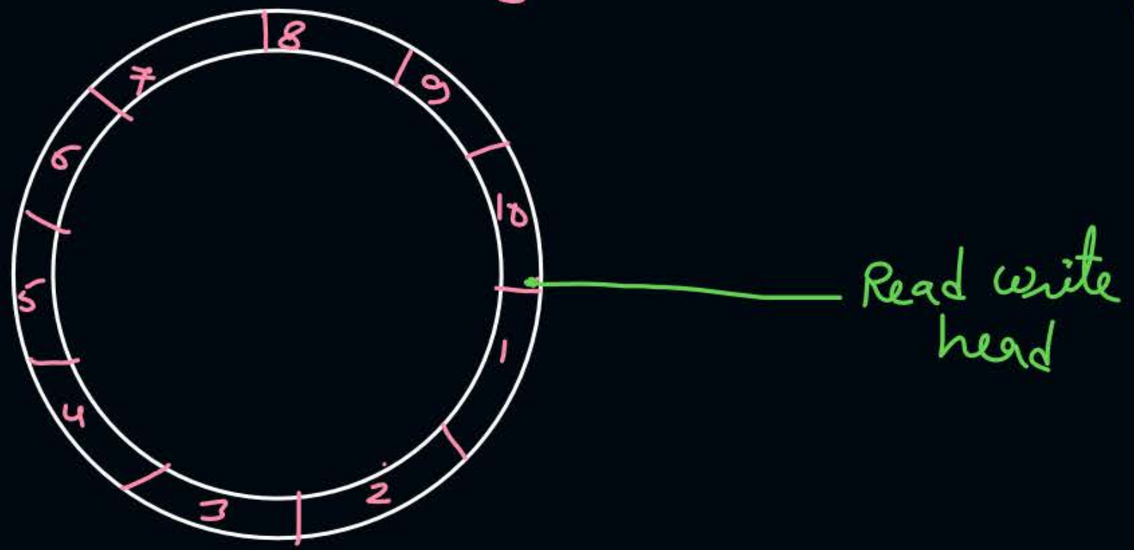
Rotational Latency: time required to rotate desired sector under R/W head

Transfer Time: Time required to read or write 1 sector

$$\text{avg. rotational latency} = \frac{1 \text{ rotation time}}{2}$$

$$1 \text{ sector transfer time} = \frac{1 \text{ rotation time}}{\text{no. of sectors per track}}$$

Track ⇒ 10 sectors



1 rotation time = 20 ms

1 sector rotation time = $\frac{20\text{ms}}{10} = 2\text{ms}$

ex 1

Target sector ⇒ 5th sector

no. of sectors to rotate = 4

rotational latency = 4 * 2 ms = 8 ms

If current and target sectors not given

⇓

Use avg. rotational latency

[NAT]

rotations per min.

#Q. Disk rotates with 1000 rpm. Find one rotation time?

for 1000 rotations = 1 min = 60 sec = 60 * 1000 msec

$$\text{for 1 rotation} = \frac{60 * 1000 \text{ msec}}{1000}$$

$$= 60 \text{ msec}$$

[NAT]

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 2K tracks per surface, 4K sectors per track and 4096 Bytes per sector. Disk rotates with 6000 rpm. Seek time is 5ms. Find disk access time?

1 rotation time $= \frac{60000}{6000} = 10ms$

$= 5ms + \frac{10ms}{2} + \frac{10ms}{4k}$

$= 10.0025\ ms$

[NAT]

400 MBPS

#Q. A disk has each track with 1k sectors each with 4KB capacity and it takes 10msec for 1 rotation. The transfer rate of the disk is?

1 track capacity = 1K * 4KB = 4MB

In 10msec, data transferred = 4MB

In 1 sec, —// — = $\frac{4MB}{10 * 10^{-3} sec}$

= 400 MB/sec

[NAT]

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 2K tracks per surface, 4K sectors per track and 4096 Bytes per sector. Disk rotates with 6000 rpm. Seek time is 5ms. Find the disk transfer rate?

1 rotation time = $\frac{60000\,ms}{6000}$

= 10ms

1 track capacity = 4K * 4KB

= 16MB

In 10ms, data = 16MB

1 sec, data = $\frac{16\,MB}{10 * 10^{-3}\,sec}$

= 1600 MB/sec



Topic : Where Disk Transfer Rate can be use?

I/O

wherever in I/O organizat^n I/O speed is needed.

[NAT]

Ans = 40 %

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 1K tracks per surface, 2K sectors per track and 2048 Bytes per sector. Disk rotates with 3000 rpm. Seek time is 10ms.

If the disk is used in cycle stealing mode of DMA, such that whenever 64-bits word is available, it will be transferred in 16ns. What is the % of time CPU is blocked?

8B

1 rotation time = $\frac{60000\,ms}{3000}$

= 20 ms

1 track capacity = 2K * 2KB

= 4MB

In 20ms, data = 4MB

1 sec, data = $\frac{4MB}{20 * 10^{-3} sec}$

= 200 MB/sec

for 200 MB, disk takes time = 1 sec

for 8 B ⟶ 1 ⟶ $= \frac{1\,sec}{200\,MB} * 8\,B$

$= \frac{8}{200}\,\mu sec$

$= \frac{8 * 1000}{200}\,ns$

$= 40\,ns$

% of time CPU blocked due to DMA $= \frac{16\,ns}{40\,ns} * 100\,ns$

$= 40\,\%$

Calculate preparatn time directly :-

for 4MB, preparation time = 20 ms

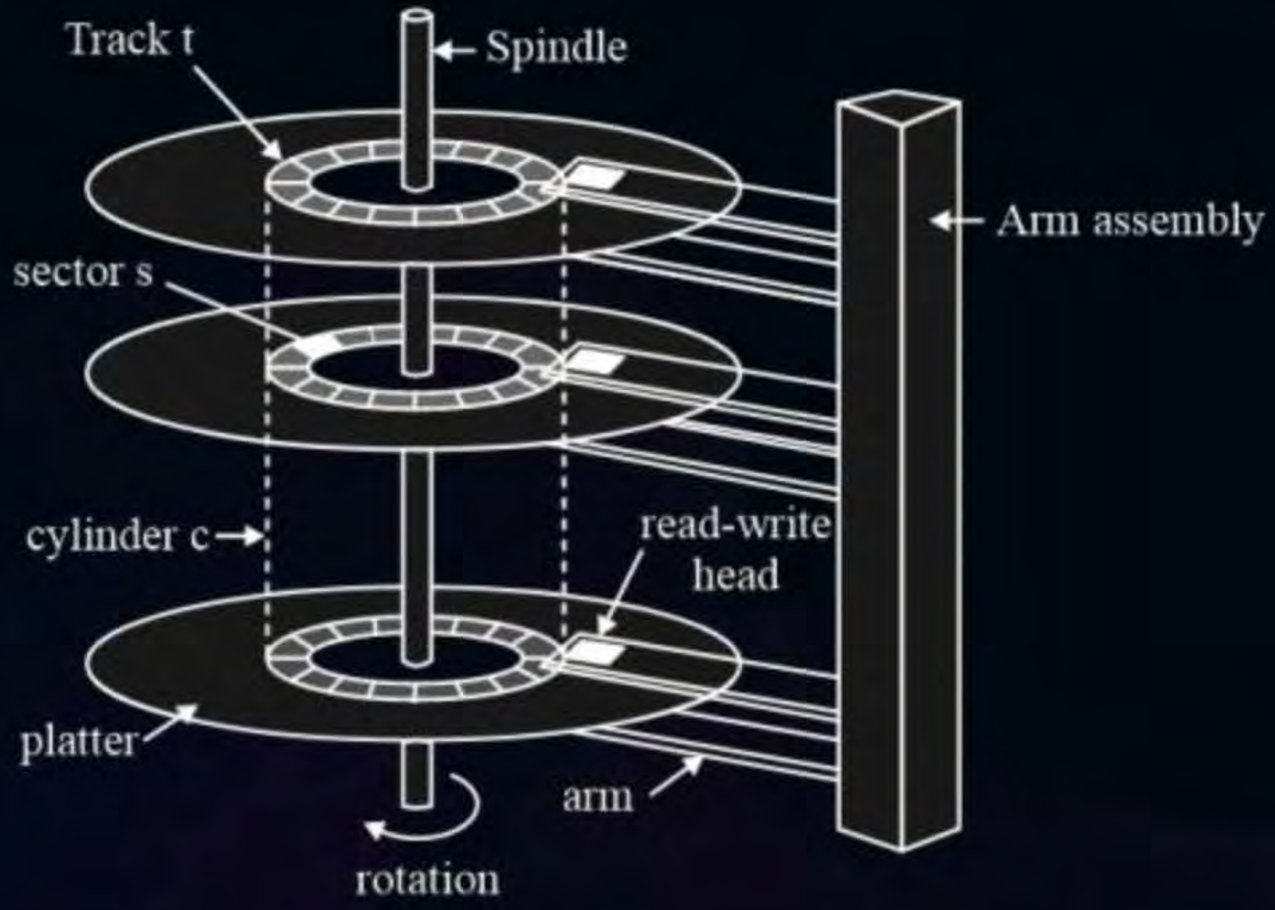
for 8B, —11— = $\frac{20\,ms}{4MB} * 8B$

= 40 ns



Topic : Multiple Sector Access Time

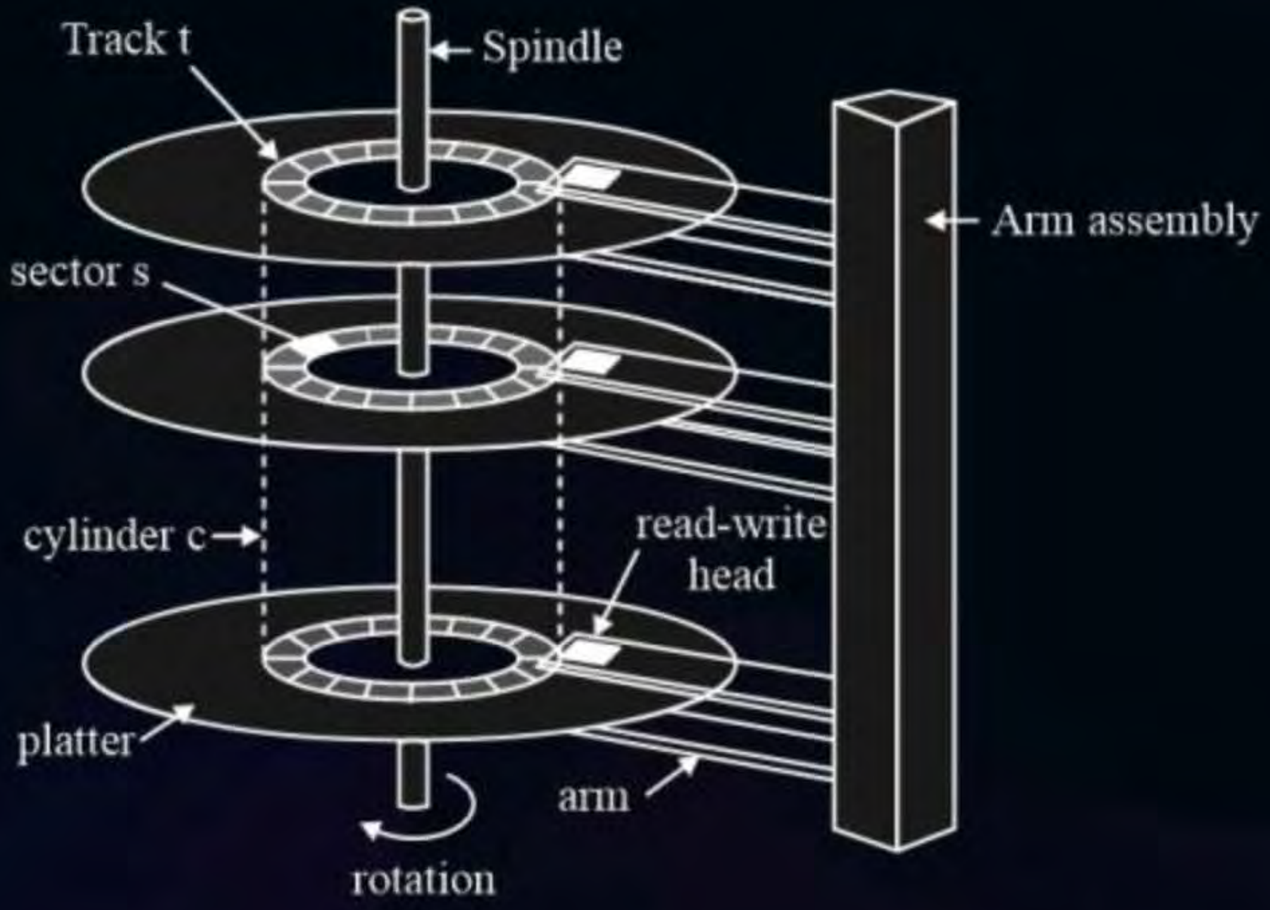
assume no. of sectors to access = n





Topic : Multiple Sectors Access Time: Sequential

In single track







Topic : Multiple Sectors Access Time: Sequential

Consider n sectors to be transferred:



n sector access time

= seek time + 1 rotational latency + n * 1 sector transfer time



Topic : Multiple Sectors Access Time: Random

Consider n sectors to be transferred:

$$n \text{ sector access time} = n * \left[\text{seek time} + \text{rotational latency} + \text{1 sector transfer time} \right]$$

[MCQ]

1 rotation time $= \frac{60000\ ms}{10000} = 6\ ms$

#Q. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is

A ✓ 14020

B 14000

C 25030

D 15000

$$= 2000 \left(4ms + \frac{6ms}{2} + \frac{6ms}{600} \right)$$

$$= 2000 * 7.01 \text{ ms}$$

$$= 14020 \text{ ms}$$



2 mins Summary

Topic Magnetic Disk

Topic Disk Capacity

Topic Disk Access Time



Happy Learning

THANK - YOU